

Laboratory Report 1

Least Squares Analysis, Curve Fitting, and Identifying Outliers

by

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Objective

The objective of this laboratory was to apply least-squares regression and curve-fitting methods to engineering data and to evaluate repeated measurements for statistically inconsistent observations using Chauvenet's criterion. Spreadsheet and manual methods were used to compare candidate correlations, determine regression equations, quantify agreement, and calculate final mean values after justified outlier screening.

Theory

Experimental measurements generally contain scatter caused by instrument resolution, measurement uncertainty, and uncontrolled variation. Regression is therefore used to determine a mathematical relationship that represents the overall behavior of a data set rather than forcing a curve through every measured point. Interpolation, by contrast, constructs a relationship that passes through specified observations. Although an interpolating equation can reproduce the supplied values exactly, it may respond poorly between observations or outside the measured range. For this reason, the assignment restricts polynomial correlations to third order or lower and requires the selection to consider both statistical agreement and the visible behavior of the data (Ranjan 2026a; Warrington 2020).

For a model that predicts a value $\hat{y}_i$ from the independent variable, the residual is the difference between the measured and predicted values:

$$r_i = y_i - \hat{y}_i \quad (1)$$

Least-squares regression selects model coefficients that minimize the sum of the squared residuals. Squaring prevents positive and negative residuals from canceling and places greater weight on larger disagreements:

$$S_r = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

For a linear relationship, the predicted dependent variable is described by a slope and intercept:

$$y = mx + b \quad (3)$$

Here, x is the independent variable, y is the dependent variable, m is the slope, and b is the vertical intercept. Minimizing the squared residuals leads to the following manual equations for the slope and intercept (Warrington 2020):

$$m = [n\sum x_i y_i - (\sum x_i)(\sum y_i)] / [n\sum x_i^2 - (\sum x_i)^2] \quad (4)$$

$$b = [(\sum x_i^2)(\sum y_i) - (\sum x_i)(\sum x_i y_i)] / [n\sum x_i^2 - (\sum x_i)^2] \quad (5)$$

The coefficient of determination measures the fraction of the observed variation represented by the model:

$$R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2] \quad (6)$$

An R^2 value closer to one normally indicates closer statistical agreement, but it does not demonstrate that a model is physically appropriate. A defensible correlation must also follow the visible pattern of the measurements, avoid unnecessary complexity, and remain consistent with the purpose of the analysis. In addition to a linear model, spreadsheet software can evaluate exponential, power, logarithmic, and polynomial relationships. A logarithmic model has the general form

$$y = a \ln(x) + b \quad (7)$$

and can represent a rapid initial increase followed by progressive flattening. Higher-order polynomials may raise R^2 while introducing curvature that is not justified by the measurements; accordingly, the course instructions prohibit orders above three and discourage selecting a model solely because it produces the largest determination coefficient (Ranjan 2026a).

When more than one independent variable is present, the same least-squares principle can be extended to a plane or hyperplane. For three independent variables, the requested model is

$$w = m_x x + m_y y + m_z z + b \quad (8)$$

Each coefficient estimates the change in w associated with its corresponding variable while the other variables are held constant. Excel's LINEST function returns the slopes, intercept, and regression statistics. Because Excel returns slopes in reverse order relative to adjacent independent-variable columns, data arranged as x , y , and z produce coefficients in the order m_z , m_y , m_x , and b ; the returned values must therefore be labeled carefully (Warrington 2020).

Chauvenet's criterion provides a systematic method for screening a set of repeated measurements. The original arithmetic mean is

$$P = (1/n) \sum_{i=1}^n P_i \quad (9)$$

and the absolute deviation of an observation from that mean is

$$d_i = |P_i - P| \quad (10)$$

The largest permissible deviation is obtained from the sample standard deviation s and the Chauvenet factor Z :

$$d_{\text{allowed}} = Zs \quad (11)$$

An observation is eligible for rejection when

$$|P_i - P| > Zs \quad (12)$$

and it is also the largest deviation in its group. Data Set 4 contains five trials at every true pressure; therefore, $n = 5$ and the spreadsheet's Chauvenet table gives $Z = 1.645$. The mean and sample standard deviation must be calculated from the original five observations, a qualifying observation may be removed once, and the criterion must not be reapplied to the reduced group (Ranjan 2026b).

Observed Data

The laboratory supplied four data sets representing four analysis cases. Tables 1 through 4 reproduce only the unprocessed measurements, as required by the course report format (Ranjan 2026c). Calculated means, regression coefficients, residuals, determination coefficients, and outlier decisions are reserved for the analysis section.

Table 1. Observed stress and strain data. The supplied strain values and corresponding measured stresses used in the curve-fitting analysis are presented.

Strain, $\epsilon \times 10^{-3}$	Stress, σ (ksi)
0.15	8.37
0.52	17.9
0.76	27.8
1.01	34.2
1.12	38.8
1.42	44.8
1.52	51.3
1.66	55.5
1.86	61.3
2.08	67.5
2.27	72.1
2.56	76.9

Strain, $\epsilon \times 10^{-3}$	Stress, σ (ksi)
2.86	83.5
3.19	88.9
3.74	94.2
4.55	100
7.27	105.3
10.1	108.3
15.4	111.4
20.3	114
25	115.4
30	116
35	116.3
40	117
45	117.2
50	117.4
55	117.7
60	118
65	119.1
70	119.5

Table 2. Reference and gauge pressures. The reference pressures and corresponding gauge-pressure readings used for the manual linear regression are presented.

Reference pressure (psig)	Gauge pressure (psig)
50	10
100	15
150	20
200	24
250	27
300	30
400	38
500	44

Table 3. Multiple-independent-variable data. The independent variables x, y, and z and corresponding dependent variable w are presented. No units were supplied for this data set.

x	y	z	w
1.5	10	0.6	0.5579193
1.5	2	0.9	0.5930470
1.5	3	1	0.5203519
1.25	9	0.7	0.5432673
1.5	1	0.7	0.7147508
1.5	7	0.9	0.5729532
1.5	8	0.9	0.5742185
1.5	6	0.8	0.5708686
1.75	2	0.4	0.6013020
1.75	8	0.2	0.5471423
1.5	10	0.2	0.5278847
1.5	3	0.9	0.5181596
1.5	6	0.2	0.5756587
1.75	7	0.2	0.5587814
1.75	5	0.2	0.6097661
1.5	8	0.8	0.5743226
1.5	7	0.8	0.5748621
1.5	1	0.6	0.7136213
1.5	2	0.8	0.5912425
1.5	6	0.7	0.5758623
1.5	3	0.8	0.5159874

Table 4. True and indicated pressures. Five repeated indicated-pressure measurements are presented for every supplied true pressure.

True pressure (psig)	Trial 1 (psig)	Trial 2 (psig)	Trial 3 (psig)	Trial 4 (psig)	Trial 5 (psig)
30	85	85	88	85	65
50	107	105	105	107	105
80	136	135	136	138	133
120	175	175	174	172	172
160	217	215	213	212	211
200	251	255	254	251	250

Analysis of the Results and Discussion

Data Set 1: Curve fitting

The 30 stress and strain observations were plotted as an XY scatter plot using unconnected markers. Linear, logarithmic, exponential, power, quadratic, and cubic least-squares correlations were evaluated. All observations were retained because the assignment explicitly states that this data set contains no outliers. Table 5 compares the determination coefficients and summarizes the model-selection decision.

Table 5. Comparison of candidate curve-fitting models. Candidate correlations were compared using statistical agreement, visible agreement with the observations, and model complexity.

Model	R ²	Decision
Linear	0.515530	Rejected as a whole-range model
Logarithmic	0.892968	Selected
Exponential	0.304208	Rejected
Power	0.597823	Rejected
Quadratic	0.693242	Rejected
Cubic	0.804996	Rejected

Figure 1 presents the original observations together with the two leading whole-range correlations. The linear model fails to represent both the steep initial increase and the later plateau. The logarithmic relationship follows those two features more closely while remaining simpler and statistically stronger than the tested polynomial alternatives.

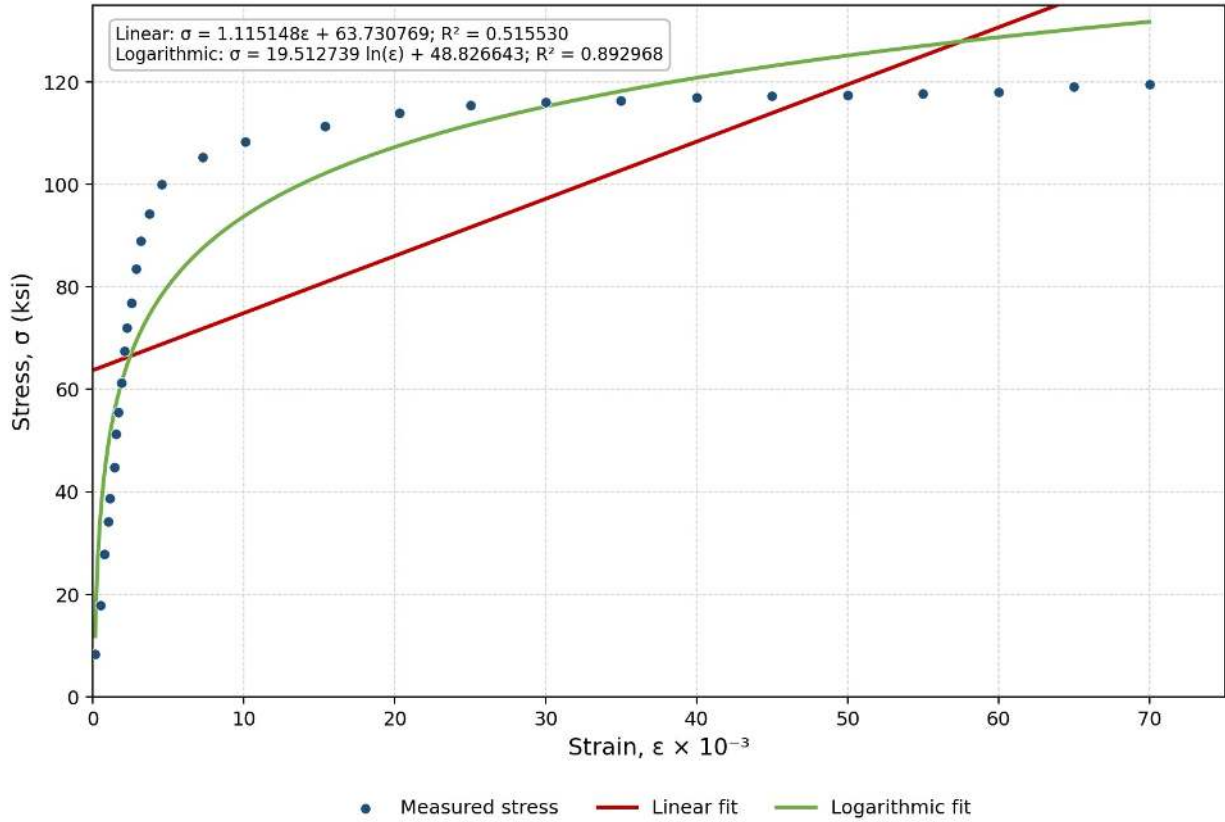


Figure 1. Stress as a function of strain with selected least-squares correlations. The measured data are shown as unconnected points, while the linear and logarithmic curves represent whole-range least-squares correlations.

The selected whole-range logarithmic model is

$$\sigma = 19.512739 \ln(\varepsilon) + 48.826643 \quad (13)$$

$$R^2 = 0.892968 \quad (14)$$

The logarithmic model produced the highest defensible R^2 among the tested candidates and reproduced the rapid initial rise and subsequent flattening without using a high-order interpolating polynomial. The cubic model produced the second-largest R^2 , but its additional coefficients and changing curvature did not improve the whole-range behavior enough to justify the added complexity. The logarithmic model was therefore selected using statistical and visual judgment rather than R^2 alone. This correlation applies only to the measured strain range because extrapolation would not be supported by the observations. Additional candidate fits are provided in Appendix A.

Data Set 2: Manual least-squares regression

The gauge pressure was treated as the dependent variable and the reference pressure as the independent variable. The calculation used the manual least-squares equations rather than a chart trendline. Table 6 shows the squared reference pressures and cross-products required for the calculation.

Table 6. Manual linear-regression calculations. The squared reference pressures and pressure cross-products used to determine the slope and intercept are presented.

x_i , reference pressure (psig)	y_i , gauge pressure (psig)	x_i^2 (psig ²)	$x_i y_i$ (psig ²)
50	10	2500	500
100	15	10000	1500
150	20	22500	3000
200	24	40000	4800
250	27	62500	6750
300	30	90000	9000
400	38	160000	15200
500	44	250000	22000
$\Sigma = 1950$	$\Sigma = 208$	$\Sigma = 637500$	$\Sigma = 62750$

With $n = 8$, $\Sigma x_i = 1950$ psig, $\Sigma y_i = 208$ psig, $\Sigma x_i^2 = 637500$ psig², and $\Sigma x_i y_i = 62750$ psig², substitution into Eqs. (4) and (5) gives

$$m = [8(62750) - (1950)(208)] / [8(637500) - (1950)^2] = 0.0742967 \quad (15)$$

$$b = [(637500)(208) - (1950)(62750)] / [8(637500) - (1950)^2] = 7.890173 \text{ psig} \quad (16)$$

The resulting calibration equation is

$$P_{\text{gauge}} = 0.0742967 P_{\text{reference}} + 7.890173 \text{ psig} \quad (17)$$

$$R^2 = 0.992545 \quad (18)$$

The determination coefficient indicates strong linear agreement over the supplied pressure range. The slope represents an increase of approximately 0.0743 psig in the gauge reading for each 1-psig increase in reference pressure, while the intercept is the regression's predicted gauge reading at zero reference pressure. The nonzero intercept is consistent with a systematic offset in the fitted relationship, although the available data do not establish its physical cause.

Data Set 3: Multiple linear regression

A multiple linear regression was calculated in Excel using $\text{=LINEST}(D3:D23,A3:C23,TRUE,TRUE)$. Because LINEST returned the slopes in z, y, x order, the coefficients were relabeled before constructing the final model. Table 7 presents the coefficients in the x, y, z order used in the equation.

Table 7. Multiple linear-regression coefficients. The coefficients determined using Excel's LINEST function are presented in the order used in the regression equation.

Quantity	Value
m_x	-0.089640
m_y	-0.011045
m_z	-0.066713
b	0.817585
R^2	0.315107

$$w = -0.089640x - 0.011045y - 0.066713z + 0.817585 \quad (19)$$

$$R^2 = 0.315107 \quad (20)$$

All three slopes are negative, and the coefficient multiplying x has the greatest magnitude. The model explains approximately 31.5% of the observed variation in w, indicating a relatively weak linear relationship. The requested analysis was linear, so the equation was retained despite the low R^2 . The complete LINEST output and observation-level residual calculations are provided in Appendix C.

Data Set 4: Chauvenet's criterion

Each true-pressure group contained five indicated-pressure observations, so $n = 5$ and $Z = 1.645$. The original mean and sample standard deviation were calculated before any removal. Table 8 compares the largest observed deviation in every group with the maximum allowed deviation Zs .

Table 8. Chauvenet screening results. Each group was evaluated using its original mean, sample standard deviation, and maximum permitted deviation.

True pressure (psig)	Original mean (psig)	Sample s (psig)	Z	Allowed deviation (psig)	Largest deviation (psig)	Decision
30	81.600	9.370	1.645	15.414	16.600	Trial 5 rejected
50	105.800	1.095	1.645	1.802	1.200	None
80	135.600	1.817	1.645	2.988	2.600	None
120	173.600	1.517	1.645	2.495	1.600	None
160	213.600	2.408	1.645	3.962	3.400	None
200	252.200	2.168	1.645	3.566	2.800	None

For the 30-psig group, the original mean was

$$P = (85 + 85 + 88 + 85 + 65) / 5 = 81.600 \text{ psig} \quad (21)$$

The sample standard deviation was

$$s = 9.370 \text{ psig} \quad (22)$$

and the maximum allowed deviation was

$$d_{\text{allowed}} = (1.645)(9.370) = 15.414 \text{ psig} \quad (23)$$

Trial 5 had the largest absolute deviation:

$$d_5 = |65 - 81.600| = 16.600 \text{ psig} \quad (24)$$

$$16.600 \text{ psig} > 15.414 \text{ psig} \quad (25)$$

Because Trial 5 both exceeded the allowed deviation and was the largest deviation, the 65-psig observation qualified for rejection. The revised mean was

$$P_{\text{final}} = (85 + 85 + 88 + 85) / 4 = 85.750 \text{ psig} \quad (26)$$

Table 9. Final pressure means following Chauvenet screening. The final mean for each pressure group is presented after removal of the single qualifying outlier.

True pressure (psig)	Original mean (psig)	Removed value (psig)	Final mean (psig)
30	81.600	65	85.750
50	105.800	None	105.800
80	135.600	None	135.600
120	173.600	None	173.600
160	213.600	None	213.600
200	252.200	None	252.200

No observation in the other five groups exceeded its corresponding allowed deviation. Chauvenet's criterion was applied only once to every original five-observation group and was not reapplied after the 65-psig value was removed.

Conclusions

The laboratory objectives were met using whole-range curve fitting, manual linear regression, multiple linear regression, and Chauvenet outlier screening. For the stress and strain data, the logarithmic correlation $\sigma = 19.512739 \ln(\epsilon) + 48.826643$ was selected with $R^2 = 0.892968$ because it represented the rapid initial rise and later flattening more closely than the other tested whole-range models without unnecessary polynomial complexity.

The manual pressure regression produced $P_{\text{gauge}} = 0.0742967P_{\text{reference}} + 7.890173$ psig with $R^2 = 0.992545$, demonstrating strong linear agreement. The multiple-variable analysis produced $w = -0.089640x - 0.011045y - 0.066713z + 0.817585$. Its R^2 of 0.315107 showed that the requested linear model explained only about 31.5% of the variation in w .

Chauvenet's criterion identified only the 65-psig value from Trial 5 at a true pressure of 30 psig as an outlier. Removing that value changed the group mean from 81.600 to 85.750 psig; no other pressure group required removal. The comparison of candidate fits also showed why model selection cannot rely on a single numerical indicator. The calculations must be considered together with the shape of the measured data and the intended use of the correlation.

References

- Ranjan, R. (2026a). "Lab 1: Least Squares Analysis, Curve Fitting, and Identifying Outliers." ENME 3070L course handout, University of Tennessee at Chattanooga, Chattanooga, TN.
- Ranjan, R. (2026b). "Chauvenet's Criterion." ENME 3070L course handout, University of Tennessee at Chattanooga, Chattanooga, TN.
- Ranjan, R. (2026c). "Fluid Mechanics Laboratory Report Format." ENME 3070L course handout, University of Tennessee at Chattanooga, Chattanooga, TN.
- Warrington, D. C. (2020). "Least Squares Analysis and Curve Fitting." Department of Mechanical Engineering, University of Tennessee at Chattanooga, Chattanooga, TN.

Appendix

Appendix A: Additional Data Set 1 correlations

Figures A1 and A2 present the four additional whole-range correlations evaluated during model selection. Each figure contains no more than two trendlines, and the original observations remain unconnected markers.

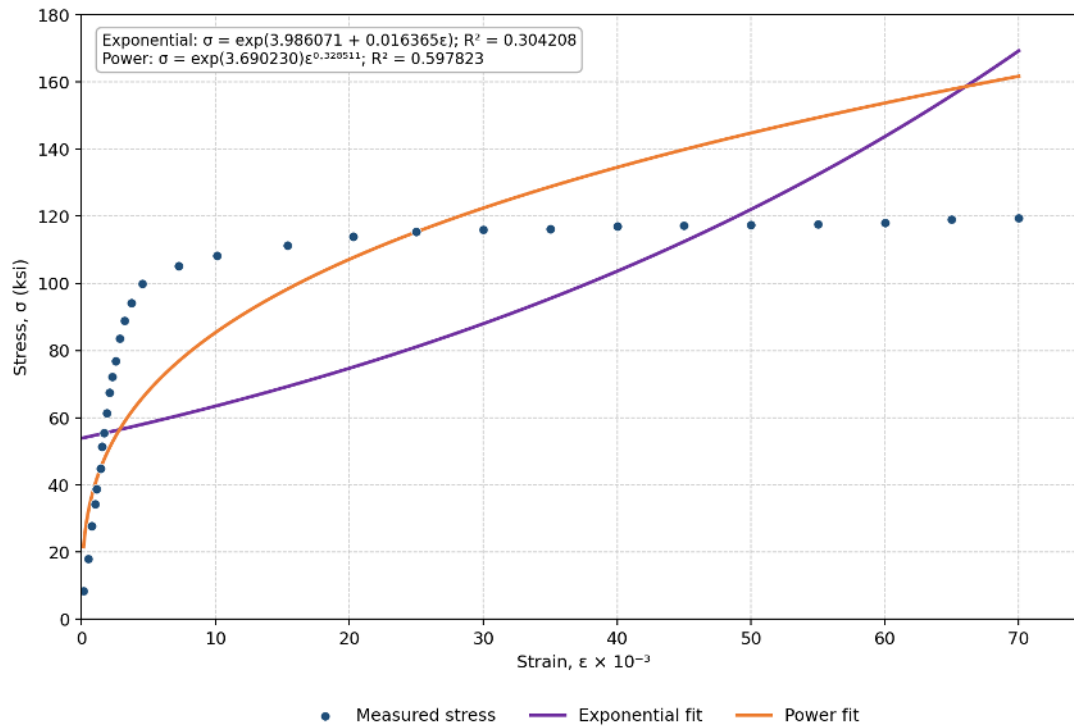


Figure A1. Exponential and power correlations. The exponential and power candidates provide weaker whole-range agreement than the selected logarithmic model.

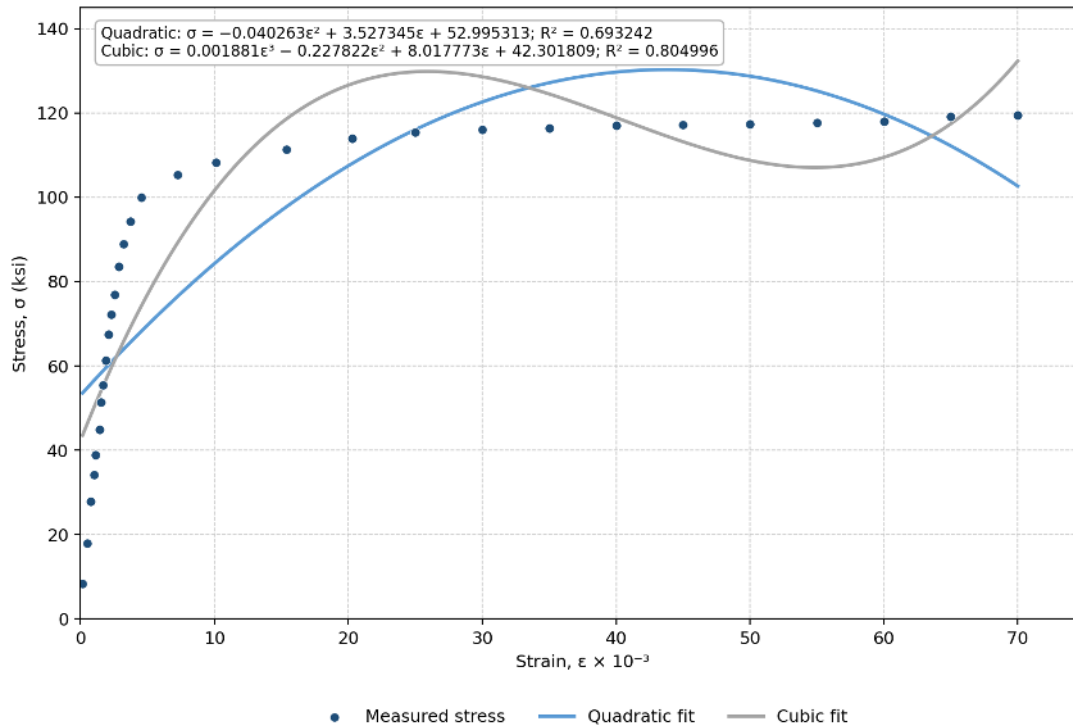


Figure A2. Quadratic and cubic correlations. The polynomial candidates improve upon the whole-range linear fit but do not represent the measured behavior as effectively as the logarithmic model.

Appendix B: Detailed manual-regression calculation table

Table B1 reproduces the complete arithmetic used to obtain the Data Set 2 sums. These spreadsheet calculations implement the manual equations without using a chart trendline.

Table B1. Detailed Data Set 2 calculations. Reference pressure, gauge pressure, squared reference pressure, and cross-products are shown with their totals.

x_i (psig)	y_i (psig)	x_i^2 (psig ²)	$x_i y_i$ (psig ²)
50	10	2500	500
100	15	10000	1500
150	20	22500	3000
200	24	40000	4800
250	27	62500	6750
300	30	90000	9000
400	38	160000	15200

x_i (psig)	y_i (psig)	x_i^2 (psig ²)	$x_i y_i$ (psig ²)
500	44	250000	22000
$\Sigma = 1950$	$\Sigma = 208$	$\Sigma = 637500$	$\Sigma = 62750$

Appendix C: Complete multiple-regression output

The live LINEST array was entered as =LINEST(\$D\$3:\$D\$23,\$A\$3:\$C\$23,TRUE,TRUE). The first row is returned in the order m_z , m_y , m_x , and b . The #N/A entries in unused cells of rows 3 through 5 are expected components of Excel's five-row output array.

Table C1. Complete LINEST output. Regression coefficients and statistics are shown in Excel's returned order.

Statistic	Column 1	Column 2	Column 3	Column 4
Coefficients	-0.066713	-0.011045	-0.089640	0.817585
Standard errors	0.052304	0.003992	0.118461	0.215043
R^2 / s_y	0.315107	0.047227	#N/A	#N/A
F / df	2.607127	17	#N/A	#N/A
SSreg / SSresid	0.017445	0.037917	#N/A	#N/A

Table C2. Multiple-regression residual verification. Predicted values, residuals, and squared residuals verify the reported determination coefficient.

Observation	Predicted w	Residual	Residual ²
1	0.532646	0.025273	0.000639
2	0.600993	-0.007946	0.000063
3	0.583277	-0.062925	0.003960
4	0.559430	-0.016163	0.000261
5	0.625381	0.089370	0.007987
6	0.545768	0.027186	0.000739
7	0.534722	0.039496	0.001560
8	0.563484	0.007385	0.000055
9	0.611939	-0.010637	0.000113
10	0.559012	-0.011869	0.000141
11	0.559331	-0.031447	0.000989
12	0.589948	-0.071788	0.005154
13	0.603512	-0.027853	0.000776
14	0.570057	-0.011275	0.000127
15	0.592147	0.017619	0.000310

Observation	Predicted w	Residual	Residual ²
16	0.541394	0.032929	0.001084
17	0.552439	0.022423	0.000503
18	0.632052	0.081569	0.006654
19	0.607664	-0.016422	0.000270
20	0.570155	0.005707	0.000033
21	0.596619	-0.080632	0.006501

Appendix D: Complete Chauvenet calculations

Table D1 shows the deviation of every trial from its original five-observation mean. The allowed deviation was calculated from Zs using $Z = 1.645$.

Table D1. Complete Chauvenet deviation calculations. All trial deviations, permitted deviations, and outlier decisions are presented.

True P	Mean	s	Zs	Dev. T1	Dev. T2	Dev. T3	Dev. T4	Dev. T5	Decision
30	81.600	9.370	15.414	3.400	3.400	6.400	3.400	16.600	Trial 5 = 65
50	105.800	1.095	1.802	1.200	0.800	0.800	1.200	0.800	None
80	135.600	1.817	2.988	0.400	0.600	0.400	2.400	2.600	None
120	173.600	1.517	2.495	1.400	1.400	0.400	1.600	1.600	None
160	213.600	2.408	3.962	3.400	1.400	0.600	1.600	2.600	None
200	252.200	2.168	3.566	1.200	2.800	1.800	1.200	2.200	None